

# Wyatt Walsh

## SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, data pipelines, and fintech developer tooling.

## EXPERIENCE

### Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.

### Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
- Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.

### Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.

### Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

## SKILLS

**MACHINE LEARNING AND GENERATIVE AI:** scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails, Semantic Retrieval

**DATA ENGINEERING AND ANALYTICS:** Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, Amazon Kinesis, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, SQLite, DuckDB, Polars, Parquet

**PROGRAMMING & SCRIPTING LANGUAGES:** Python, SQL, TypeScript, JavaScript, Go, Bash

**CLOUD INFRASTRUCTURE AND DEVOPS:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, Serverless Architecture, AWS Lambda, ECS, EKS, Vercel

## PROJECTS

**AI Agent Harness Configs** - Python/TypeScript MCP workflow catalog with 40+ reusable skills, FastMCP configs, and cross-runtime agent automation.

- Built 40+ instruction and skill packs plus harness configs across Claude Code, Codex, GitHub Copilot, Cursor, and related IDE agents.
- Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

**NBA Basketball Database** - Python warehouse using Pandas, Polars, SQLAlchemy, DuckDB, and Pandera across 65K+ games, 13M+ events, and Parquet/SQLite exports.

- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

**ProxyWhirl** - Python/SQLite proxy toolkit with nine rotation strategies, 100+ sources, PyPI distribution, and self-healing health checks.

- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
- Built self-healing validation for 100+ proxy sources with circuit breakers, auto-ejection, and recovery promotion.

**Personal Website: w4w.dev** - Next.js/React/TypeScript publishing system with Tailwind CSS, MDX, durable content, SEO, and polished public UI.

- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
- Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

## EDUCATION & CERTIFICATIONS

### Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)  
Aug 2014 - Dec 2020

### High School Diploma

The Hotchkiss School  
Aug 2012 - May 2014

### AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

### IBM Data Science Professional Certificate

IBM | Feb 2021